

EDUCATION	Iowa State University Ph.D. Student, Computer Science, GPA 3.83/4.0 Advisor: Dr. Yan-Bin Jia Ames, IA, USA Aug. 2016 - <i>present</i>
	University of Science and Technology of China B. S., Modern Physics, Graduated with honors Hefei, China Jun. 2016
PROFESSIONAL EXPERIENCE	Iowa State University Research Assistant , Robotics Lab Ames, IA, USA Aug. 2016 - <i>present</i>
	- Developed a robotic maneuver for picking up a kitchen knife from the table by a robotic arm and anthropomorphic hand using pivoting and finger gaiting. - Proposed a sequence of hybrid control policies for a robotic arm-hand system performing the smooth food cutting. - Investigated the contact kinematics and dynamics for manipulating an object via a robotic hand with coupled joints; Proposed a control algorithm to re-position and re-orient an object within the hand.
	Intel Open Souce Technology Center Intern , Linux Kernel and Virtualization Beijing, China Jun. 2018 - Sep. 2018
	- Implemented a CLI tool for spawning, running, and managing containers on the Linux platform according to the OCI specification. - Delivered a presentation on state-of-the-art GPU virtualization technologies.
	DJI Intern , Navigation and Vision Algorithm Shenzhen, China Mar. 2016 - Jul. 2016
	Prototyped a UAV obstacle sensing and tracking subsystem that utilizes long-range stereo vision, multi-sensor fusion, and state estimation.
	Conference Reviewer
	- IEEE/RSJ International Conference on Robotics and Automation (ICRA) - IEEE International Conference on Intelligent Robots and Systems (IROS)
PUBLICATIONS	Y. Xue, and Y.-B. Jia. <i>Gripping a kitchen knife on the cutting board</i>. In 2020 International Conference on ntelligent Robots and Systems (IROS). IEEE, 2020. X. Mu, Y. Xue, and Y.-B. Jia. <i>Robotic cutting: Mechanics and control of knife motion</i>. In 2019 International Conference on Robotics and Automation (ICRA), pp. 3066-3072. IEEE, 2019. (Highlighted by IEEE Spectrum Robotics News and The Spoon News) - Y.-B. Jia, and Y. Xue. <i>Dexterous manipulation by two fingers with coupled joints</i>. In IEEE International Conference on Robotics and Automation (ICRA), pp. 3172-3179. IEEE, 2018. X. Mu, Y. Xue, and Y.-B. Jia. <i>Robotic cutting: Mechanics and knife control</i>. Submitted to IEEE Transactions on Robotics (T-RO).
TECHNICAL SKILLS	Programming: C/C++ and Rust for real-time robotic control and simulation; Python, Matlab as primary development languages; V+ for Omron Adept robot programming. Other experience includes Java, Go, Verilog (Xilinx Vivado), JavaScript, C#. Robotics: ROS, Gazebo for robot controller prototype; MuJoCo, Bullet, Matlab for robotic dynamics simulation; OpenCV for computer vision; OMPL for path planning. Systems and Tools: Linux Kernel, CPU virtualization, Tensorflow, Android, Git, CI workflow, AWS.

AWARDS	Robert Stewart Early Research Recognition Award, Iowa State University	2018
	3 rd Place as team leader, Iowa State University CS Hackathon	2017
	Meritorious Winner, Mathematical Contest in Modeling	2016
	Gold Award as team leader, International Genetically Engineered Machine Competition (iGEM)	2015
	1 st Place, Thesis Competition on Electromagnetism, Univ. Sci. Tech. China	2013
	Gold Award, Outstanding Student Scholarship, Univ. Sci. Tech. China	2012
TEACHING	Teaching Assistant, Iowa State University	Aug. 2016 - <i>present</i>
	1. Com S 228 - Data Structure	
	<i>Developed a software for automatic homework grading.</i>	
	2. Com S 327 - Advanced Programming Techniques (C/C++)	
	3. Com S 352 - Introduction to Operating System	
	4. Com S 572 - Artificial Intelligence	
	5. Com S 576 - Motion Strategy Algorithms	
SELECTED PROJECTS	6. Com S 577 - Problem Solving Techniques for Applied Computer Science	
	<i>Implemented high quality C++ solution codes for all coding projects.</i>	
	1. nurbs-rs : A pure Rust, self-contained, object-oriented B-Spline and NURBS library.	
	Available at https://github.com/4x7y/nurbs-rs .	
	2. Realtime Linux Patch : A realtime Linux patch that enables kernel memory mapping function kmap preserving mapping slots for realtime tasks and kill lowest priority task to release resource if necessary.	
	3. Container Runner : A lightweight container runner for the Linux platform implemented in Go and C. Features including resource management (cgroups), container network (ip-netns), isolated filesystem (chroot, aufs, tmpfs) and etc.	
	Available at https://github.com/4x7y/mydocker .	
	4. Multi-Camera SLAM : A multi-camera SLAM ROS module based on RTABMap. 3D vision sensors can work collaboratively to build a map by merging frames in a centre node.	
	5. Mini CPU Design : A single cycle mini CPU with 2 stage pipeline. Assembly code and Verilog code co-design and co-verification. Real testing on a Xilinx FPGA board for simple applications (processing instructions from a PC through serial port).	
	Available at https://github.com/4x7y/Mini-CPU	